

AWS Cloud Practitioner Cheat Sheet Table

Category	Service Name	What it does (Exam Definition)	Key Exam Hint / Differentiator	Real-World Example
Compute	EC2	Virtual servers (Instances) in the cloud.	Pay-as-you-go. Combine with Auto Scaling and ELB . Not serverless (you manage the OS).	Moving your company's web server to the cloud.
Compute	Lambda	Run code without provisioning servers.	Serverless. Event-driven. Charges per request and compute time (milliseconds).	Automatically resize images when a user uploads them to S3.
Compute	Elastic Beanstalk	PaaS (Platform as a Service). Deploy code, AWS handles the underlying EC2, scaling, and load balancing.	You just upload your code (Java, Python, etc.). Free (you pay for the underlying EC2/S3).	Deploying a web app without needing to click "Launch Instance" manually.
Compute	ECS / EKS	Container orchestration (Docker). ECS = AWS Native; EKS = Kubernetes.	Use Fargate for serverless containers (no EC2 management).	Running a microservices architecture using Docker containers.
Storage	S3	Object storage (files, images, videos).	Unlimited storage. Serverless. Global namespace (buckets are unique globally).	Storing website images, backups, or data lakes.

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Storage	S3 Glacier	Cheap, long-term archival storage.	Retrieval times: Instant, Minutes, or Hours (cheaper = slower). Not for frequent access.	Storing 7-year financial records for compliance.
Storage	EBS	Block storage (like a virtual hard drive) for EC2 .	Must be in the same AZ as the EC2. Persists even if EC2 stops.	Attaching a 1TB drive to a database server.
Storage	EFS	Network file system (NFS). Shared storage.	Can be mounted across multiple EC2 instances and multiple AZs . Serverless .	Hosting a WordPress site where multiple EC2 instances share the same media folder.
Databases	RDS	Relational Database (SQL): MySQL, PostgreSQL, Oracle, etc.	Managed (AWS patches OS/backups). Multi-AZ for DR, Read Replicas for performance.	Moving your CRM's SQL database to the cloud.
Databases	DynamoDB	NoSQL Key-Value / Document database.	Serverless . Millisecond latency. Scales automatically.	Storing user session data for a gaming app or product catalogs.
Databases	Redshift	Data Warehousing (Analytics).	Used for Big Data and OLAP (Online Analytical Processing). Columnar storage.	Analyzing 5 years of sales data to find business trends.

Category	Service Name	What it does (Exam Definition)	Key Exam Hint / Differentiator	Real-World Example
Network	VPC	Virtual Private Cloud (your isolated network in AWS).	Contains Subnets (Public vs Private), Route Tables , and NACLs (Stateless firewalls).	Creating a secure, isolated network for your servers.
Network	CloudFront	Global Content Delivery Network (CDN).	Uses Edge Locations . Caches static content. Integrates with S3 and EC2 .	Speeding up video delivery to users in Europe by caching it locally.
Network	Route 53	DNS (Domain Name System) service.	Translates domain names (e.g., example.com) to IP addresses. Supports routing policies .	Pointing " mycoolwebsite.com " to your EC2 server's IP.
Network	API Gateway	Creates REST/WebSocket APIs.	Serverless front-door for Lambda. Handles throttling, authentication, and caching.	A mobile app talks to API Gateway, which triggers a Lambda function to fetch user data.
Security & Identity	IAM	Identity and Access Management (Users, Groups, Roles, Policies).	Global (not region-specific). Least privilege principle . Not used for application auth (use Cognito).	Giving the HR team read-only access to S3 buckets containing resumes.
Security & Identity	Cognito	User authentication, sign-up, and sign-in for web/mobile apps.	Provides Federated Identity (Login with Google/Facebook) and temporary AWS credentials.	Adding "Login with Google" to a mobile app to allow users to upload files to S3.

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Security & Identity	WAF / Shield	Web Application Firewall (Protects against SQL injection, XSS) & DDoS protection.	Shield Standard = Free (automatic). Shield Advanced = Paid (for large DDoS attacks).	Blocking malicious bot traffic trying to brute-force your login page.
Security & Identity	KMS	Key Management Service (Encryption).	Manage Customer Master Keys (CMKs) . Integrates with S3, EBS, RDS. Hardware Security Module (HSM) available.	Encrypting data inside an S3 bucket using a custom encryption key.
Integration & Messaging	SQS	Simple Queue Service (Decoupling applications).	Pull-based. Serverless. Used for buffering requests. Standard (high throughput) vs FIFO (strict order).	Sending web requests to a queue to be processed by a Lambda function when the system has capacity.
Integration & Messaging	SNS	Simple Notification Service (Pub/Sub messaging).	Push-based. Sends messages to subscribers (Email, SMS, Lambda, SQS).	Sending an email/SMS alert to a SysAdmin when a server's CPU reaches 90%.
Cost & Management	AWS Pricing	Pay-as-you-go , Reserved Instances (1/3yr discount), Spot Instances (up to 90% off but interruptible).	Savings Plan = Commit to compute usage. Free Tier = 12-month free (EC2) + always free (Lambda/S3).	Saving costs on a 24/7 production database using a 1-year Reserved Instance.

Category	Service Name	What it does (Exam Definition)	Key Exam Hint / Differentiator	Real-World Example
Cost & Management	AWS Budgets	Set custom budgets to track cost and usage.	Alerts you when you exceed thresholds. Forecasts future spending.	Getting an alert if your monthly bill is projected to exceed \$500.
Cost & Management	AWS Cost Explorer	Visualize, understand, and manage your AWS costs over time.	Uses Filters (Service, Region, Tag). Helps find cost drivers.	Checking why your data transfer costs spiked last month.
Cost & Management	Trusted Advisor	Automated cloud best practice inspector (Performance, Security, Cost, Fault Tolerance).	Free Tier = 7 core checks. Business/Enterprise = Full checks.	Getting a recommendation to delete unused EC2 instances to save money.
Migration & Transfer	Snowball Edge	Physical data transport device (Petabytes of data).	Used when the internet is too slow to transfer data. Edge computing capabilities.	Moving 100TB of video footage to AWS from a studio that has slow upload speeds.
Migration & Transfer	DataSync	Online data transfer service.	Syncs on-premise data to AWS or between AWS services (EBS to S3).	Automatically backing up local file servers to AWS every night.
Management & Governance	CloudFormation	Infrastructure as Code (IaC).	Uses templates (YAML/JSON) to provision AWS resources automatically. Declarative (you define the "what").	Spinning up 50 identical servers, databases, and networks with one script.

Category	Service Name	What it does (Exam Definition)	Key Exam Hint / Differentiator	Real-World Example
Management & Governance	CloudTrail	Who did what , when , and where ? (Audit log of all API calls).	Investigation and Security Auditing . Logs are stored in S3.	Figuring out who deleted an important S3 bucket yesterday at 3 PM.
Management & Governance	CloudWatch	Monitoring performance metrics (CPU, Memory) and Logs.	Performance monitoring. Alarms trigger actions (e.g., Auto Scaling).	Setting an alarm to trigger an Auto Scaling policy when CPU > 80%.

5 Critical "Trick" Questions for the Exam

Knowing the definitions is step one. Here is how AWS tries to trick you on the exam:

1. High Availability vs. Disaster Recovery:

- *Trick:* If the question mentions "power outage in us-east-1," the answer is usually **Multi-AZ** (which keeps data sync in real-time).
- *Wrong Answer:* "Back up to S3" (that is Disaster Recovery, meaning you lose data if the primary fails).

2. Scalability:

- *Trick:* If the question mentions "increasing the size of the instance," that is **Vertical Scaling** (EC2 size upgrade).
- *Wrong Answer:* Don't choose "Auto Scaling" unless the scenario mentions adding **more instances** (Horizontal Scaling).

3. S3 Standard vs. Glacier:

- *Trick:* If the scenario says "Access frequently" or "Website hosting," choose **S3 Standard**.
- *Wrong Answer:* If the scenario says "Regulatory archives accessed once a year," choose **Glacier Deep Archive**, not S3 Standard.

4. CloudTrail vs. CloudWatch:

- *Trick:* If the scenario says "Someone changed a security group setting," the answer is **CloudTrail** (Audit of API calls).
- *Wrong Answer:* **CloudWatch** is for CPU/Memory performance, not user actions.

5. SQS vs. SNS:

- *Trick:* If the scenario says "Decouple two applications" or "Buffer requests," the answer is **SQS** (Queue).
- *Wrong Answer:* If the scenario says "Send an email/sms to users," the answer is **SNS** (Notification).

Quick Architecture Scenarios (Memorize These)

Scenario	Best AWS Solution
Host a static website (HTML, CSS, JS) with low cost.	S3 (Static Website Hosting) + CloudFront .
Host a dynamic website with a database and autoscaling.	EC2 (in ASG) + RDS + ELB .
Run code without managing servers (e.g., image processing).	Lambda (triggered by S3 event).
Store 10TB of data cheaply that is accessed once a year.	S3 Glacier Deep Archive .
Ensure a database does not fail if one AZ goes down.	RDS Multi-AZ (Synchronous replication to a standby).
Improve database read performance for heavy traffic.	RDS Read Replicas (Asynchronous replication).
Send an alert to the SysAdmin's phone.	SNS (push notification).
Process a queue of orders one by one in strict order.	SQS FIFO .

Pro Tip for the exam: If you don't know the answer, look for keywords in the question.

- "Cost-effective" + "No server management" = **Serverless** (Lambda/DynamoDB/S3).
- "Global" + "Low latency" = **CloudFront** (CDN) or **Global Accelerator**.
- "Hybrid" (On-prem + Cloud) = **Direct Connect** or **Storage Gateway**.

Trick 1: The "EBS" Multi-AZ Trap

- **The Scenario:** A company has a critical application on an EC2 instance with an EBS volume. They need **high availability** so that if one data center fails, the application continues to work.
 - **The Trick Answer (Wrong):** "Take a snapshot of the EBS volume and copy it to another Region."
 - **The Correct Answer: Create an AMI (Amazon Machine Image)** and launch the EC2 instance into an **Auto Scaling Group** across **multiple Availability Zones**, or use an **EBS Replication** solution.
 - **Exam Rule: EBS Snapshots are for Disaster Recovery (Backup), NOT High Availability.** HA means it keeps running during a failure; Snapshots take minutes/hours to restore, meaning *downtime*.
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Trick 2: The "S3 Bucket" Global vs. Regional Trap

- **The Scenario:** You create an S3 bucket named my-company-data. You try to create a second bucket with the exact same name in a different AWS Region, and it fails.
 - **The Trick Answer (Wrong):** "S3 buckets are Region-specific, so you cannot create the same name in different Regions."
 - **The Correct Answer: S3 bucket names are GLOBALLY unique across all AWS accounts and all Regions.**
 - **Exam Rule:** While S3 *data* stays in the Region you choose, the *namespace* (bucket name) is global. If someone in Australia has my-data, you cannot use that name in the US.
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Trick 3: The "Security Group" Stateful Trap

- **The Scenario:** You configure a Security Group (firewall for EC2) with an **Inbound Rule** allowing traffic on port 80 (HTTP) from the internet. You set **Outbound Rules** to "Deny All."
 - **The Trick Answer (Wrong):** "The web server will not respond because Outbound is blocked."
 - **The Correct Answer: The web server WILL respond.**
 - **Exam Rule: Security Groups are STATEFUL.** If you allow inbound traffic, the return/response traffic is automatically allowed, *regardless* of outbound rules. (Note: **NACLs** are Stateless and would block the response).
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Trick 4: The "Reserved Instance" Region Trap

- **The Scenario:** You purchase a 1-year **Regional Reserved Instance** for a t2.micro in us-east-1. You launch a t2.micro EC2 instance in us-west-2.
 - **The Trick Answer (Wrong):** "The Reserved Instance discount will automatically apply to the us-west-2 instance because it's the same family."
 - **The Correct Answer:** The discount does **NOT** apply.
 - **Exam Rule:** **Regional Reserved Instances** apply to that *specific Region* (and AZ, depending on type). A Zonal RI applies to a specific AZ. They do not cross Regions. You must buy an RI in us-west-2 to get the discount there.
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Trick 5: The "Spot Instance" Termination Trap

- **The Scenario:** You launch a Spot Instance to process a batch job. AWS needs the capacity back and terminates your Spot Instance with a 2-minute warning.
 - **The Trick Answer (Wrong):** "The Spot Instance will be stopped and saved, and you can restart it later."
 - **The Correct Answer:** The Spot Instance is **TERMINATED (deleted)**, not stopped.
 - **Exam Rule:** Spot Instances are **interruptible** and are **terminated** when reclaimed. You lose the data on the local instance store. Only *On-Demand* or *Reserved* instances can be "Stopped" and "Started."
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Trick 6: The "S3 Storage Class" Lifecycle Trap

- **The Scenario:** You upload a file to S3 Standard. You set a Lifecycle Rule to move objects to **S3 Glacier** after 30 days, and delete them after 90 days.
 - **The Trick Answer (Wrong):** "The file will be automatically deleted 90 days after it was uploaded."
 - **The Correct Answer:** The file will be deleted **120 days after upload** (30 days in Standard + 90 days in Glacier).
 - **Exam Rule:** Lifecycle transitions **add time**. The clock resets at each transition. AWS exams love this math.
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Trick 7: The "ELB" Health Check Trap

- **The Scenario:** An Application Load Balancer (ALB) sits in front of EC2 instances. The EC2 instances are running perfectly (CPU is fine), but the ALB keeps marking them as "Unhealthy" and terminating them.
- **The Trick Answer (Wrong):** "The EC2 instances must be out of memory."
- **The Correct Answer:** The **ALB Health Check is failing** (e.g., it's pinging /health.html on the instance, and that specific page doesn't exist or returns a 404 error).

- **Exam Rule:** Load Balancers do NOT care about CPU/Memory for health checks. They only care about the **HTTP response code** (200 OK) from the specific health-check endpoint you configured.
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Trick 8: The "CloudFront" Origin Trap

- **The Scenario:** You have an S3 bucket hosting a static website in us-east-1. You set up **CloudFront** to cache the content globally. You update an image in the S3 bucket, but users are still seeing the old image.
 - **The Trick Answer (Wrong):** "You need to wait for the S3 bucket to replicate the file."
 - **The Correct Answer:** You need to **invalidate the cache in CloudFront** (or wait for the TTL - Time To Live - to expire).
 - **Exam Rule:** CloudFront **caches** content at Edge Locations. Updating the "Origin" (S3) does not instantly update the cache. You must force a refresh or use versioned filenames (e.g., image_v2.png).
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Trick 9: The "IAM" Cross-Account Trap

- **The Scenario:** You have two AWS accounts: Account A (has an S3 bucket) and Account B (has EC2). You attach a policy to a user in Account B giving them s3:GetObject permission on Account A's bucket.
 - **The Trick Answer (Wrong):** "The user in Account B can now read the S3 bucket."
 - **The Correct Answer:** The user cannot access it.
 - **Exam Rule:** For cross-account access, **BOTH** sides need a policy. Account A must have a **Bucket Policy** explicitly allowing Account B to access it. The user policy in Account B is not enough on its own.
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Trick 10: The "Snowball" vs "DataSync" Trap

- **The Scenario:** A company has a 500 GB database on-premises. They have a fast, stable 1 Gbps internet connection. They need to migrate this database to AWS RDS **once**.
 - **The Trick Answer (Wrong):** "Use AWS Snowball Edge to physically ship the database."
 - **The Correct Answer:** Use **AWS DataSync** or the **AWS Schema Conversion Tool / DMS** (Database Migration Service) over the internet.
 - **Exam Rule:** **Snowball** is for **Petabytes** (massive data) or situations with **slow/poor internet** (ships, remote locations). 500GB over a fast internet connection is cheaper and faster than shipping a physical device.
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Quick "Keyword" Cheat Sheet to Spot Tricks:

Keyword in Question	What AWS Actually Means (The Trap)
" Stop a Spot Instance"	Impossible. Spot instances can only be Terminated or Interrupted .
" Low latency " + " Global "	The answer is almost always CloudFront or Global Accelerator , not EC2 in a single region.
" Copy an EBS Snapshot"	This is for Disaster Recovery (takes minutes/hours). If they say " No downtime ", choose Multi-AZ instead.
" Outbound traffic blocked"	If it's a Security Group , traffic still works (Stateful). If it's a NACL , traffic is blocked (Stateless).
" S3 Standard-IA "	This is for <i>infrequent</i> access but rapid retrieval . If they say " Archival ", you must pick Glacier .
" Build a custom VPC "	They are likely testing if you know that Subnets are tied to one AZ (you cannot span subnets across AZs).
" Automatically scale based on CPU"	The answer is Auto Scaling Group . (Not Elastic Beanstalk, which just deploys code).